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Influences of Continuous Sitting and Psychosocial Stress on Low Back Kinematics, Discomfort, and Localized Muscle Fatigue During Unsupported Sitting Activities

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Abstract

Continuous seated postures may increase the risk of adverse health outcomes such as low-back pain, and this risk may be influenced by several modifying influences. In the present study, we aimed to quantify the effects of continuous sitting and psychosocial stress under an unsupported sitting condition. Fourteen participants completed continuous, 40-min. periods of computer-based tasks, involving both low and higher levels of psychosocial stress, while using a laptop computer without a desk. Continuous sitting significantly increased perceived discomfort (particularly in the upper and lower back), trunk flexion, and metrics of localized muscle fatigue. A higher level of psychosocial stress increased estimated lumbosacral compression forces (by ~12%). Only weak correlations were found between subjective and objective measures, while various fatigue metrics showed a good level of correspondence with each other. These results could support the future evaluation or design of diverse seated work configurations.

Keywords: Continuous sitting; Psychosocial stress; Perceived discomfort; Muscle Fatigue; Lumbosacral kinematics; Spine Biomechanics; Laptop Computer Use; Unsupported Sitting

Practitioner Summary

Continuous, 40 min. periods of unsupported sitting had broad impacts on subjective and objective outcomes, including discomfort, postures, spine loads, and localized muscle fatigue, while psychosocial stress only had a substantial influence on lumbosacral compression. These results extend our understanding of sitting behaviors and provide information for designing future sitting environments.

Nomenclature			
LBP	Low Back Pain	MPF	Median Power Frequency (Hz)
EMG	Electromyography	PCA	Principal Component Analysis
LS	Low Psychosocial Stress	EPC1	Dominant principal component of EMG
HS	High Psychosocial Stress	PPC1	Dominant principal component of perceived discomfort scores
FR	Flexion-Relaxation	ANOVA	Analysis of Variance
SR	Stimulation Response	MANOVA	Multivariate Analysis of Variance
MVC	Maximal Voluntary Muscle Contraction		
F _{ANT}	Anterior-Posterior Shear Force (N)	Left (L) and Right (R) sides for the following muscle groups:	
F _{COMP}	Compression Force (N)	EL	Erector Spinae Lumbar Group
F _{LAT}	Lateral Shear Force (N)	ET	Erector Spinae Thoracic Group
M _{LAT}	Lateral Bending Moment (Nm)	MF	Multifidus
M _{AXIAL}	Axial Rotation Moment (Nm)	IO	Internal Oblique
M _{FLEX}	Flexion Moment (Nm)	EO	External Oblique
PD	Perceived Discomfort	RA	Rectus Abdominis

1 Introduction

Sedentary behavior, especially prolonged continuous sitting, presents emerging health risks in both occupational and non-occupational settings (Owen et al., 2014). Recent estimates indicate that many adults maintain seated postures in over 90% of their leisure time (Chau et al., 2012) and over half of their working hours (Thorp et al., 2012; Bronwyn et al., 2015). Furthermore, individuals who work mainly in a seated posture are more likely to experience or aggravate low back pain (LBP: NIOSH, 1997; Dankaerts et al., 2006; O'Sullivan et al., 2006; Womersley and May, 2006). Sitting behaviors can also be affected by diverse job-related factors, such as the working

environment (Magnusson and Pope, 1998; Delleman et al., 2004), seated work duration (Keyserling, 1991; Pitman and Ntuen, 1996; Stricker et al., 2003), or psychological task stress (Ellegast et al., 2007; Taib and Yun, 2014).

Due to the flexibility of using mobile devices, such as smartphones, laptops, or tablets, many users choose to use these portable devices in situations without a desk. In addition, many places involve situations in which unsupported seating is all that is available (i.e., non-desk conditions with or without a backrest). For example, flexible library and activity-based, open-plan office concepts turn many workplaces into a more flexible environment, but with no designated workstation assigned to any user. To maximize occupancy, unsupported seating environments are also common solutions provided in public areas, such as airport or coffee shops. These environments often provide a mix of benches, chairs, booths, and lounge areas (Figure 1) for various collaborative activities, but they can also require individuals to work in unsupported seating conditions (Rianne et al. 2011, Asher 2017). The current study employed particular scenarios involving unsupported sitting, simulating continuous working conditions with limited in-chair movements. Using mobile devices under such sitting conditions could increase exposure to postures that may increase the risk of musculoskeletal disorders (Kim 2015, Yassierli and Juraida 2016).



Figure 1. Examples of activity-based, open-plan office and library environments.

Prolonged sitting often results in discomfort and/or pain in the lower back (Fenety et al., 2000; Beach et al., 2005; Dunk and Callaghan, 2005; Gregory et al., 2006).

However, the effects of a single continuous period of sitting on worker performance are not well studied, and there remains a lack of evidence quantifying the development of discomfort and associated risks. In addition, high mental workloads and stressful working schedules are often characteristics of modern jobs that require continuous sitting. Psychosocial stress describes non-physical aspects of work, such as time pressure and high information processing demands, each of which has been found to contribute to the risks of developing work-related musculoskeletal disorders (Bongers et al. 1993, Hartvigsen et al. 2004). More specifically, psychosocial stress has been related to the development of work-related musculoskeletal disorders in upper extremities (Hughes et al., 2007; Thiese, et al., 2015), shoulder (Blangsted et al., 2004; Mehta and Agnew, 2012), and back (Hartvigsen et al., 2004; Marras et al., 2000; Kato, et al., 2013; Taib and Yun, 2014). However, the combined effects of psychosocial stress and continuous sitting on LBP risks have not been quantified.

There is also some inconsistency in reported evidence regarding the relationship between these factors and the risk of sitting-related LBP (Leboeuf-Yde, 2004; Womersley and May, 2006; Lis et al., 2007), which may be due, at least in part, to the insensitivity of the evaluation methods employed. The effects of continuous sitting on lumbosacral loads may be caused by time-dependent changes in muscle fatigability and/or viscoelastic behavior of passive trunk components (Bigland-Ritchie et al., 1995; McLean et al., 1997; Keller and Nathan, 1999; Beach et al., 2005). Localized muscle fatigue has been measured via muscle EMG to assess the effects of continuous sitting on the lumbosacral region (McLean et al., 1997; Seghers et al., 2003), yet EMG-based assessments of muscle fatigue can be relatively less effective while muscle activities are less than 15% of maximal muscle capacity (Öberg et al., 1992; Öberg et al., 1994). Other more sensitive methods, such as based on external muscle stimulation, provide an alternative approach to reliably detect muscle fatigue during low-level muscle contractions (Johnson, 1998; Jia et al., 2015; Jeremy et al., 2015; Garcia, et al., 2015). An accurate estimation of lumbosacral loads during prolonged sitting is also reliant on appropriately modeling viscoelastic tissue behaviors. No existing study, to our knowledge, has quantified potential time-dependent changes in lumbosacral loads resulted from continuous sitting, with consideration of both the active (via muscle fatigue) and passive (via viscoelastic behaviors) contributions of the lumbar tissue.

Therefore, this study was designed to determine the effects of continuous sitting and psychosocial stress under an unsupported sitting condition, using both subjective (discomfort) and objective (lumbosacral kinematics and kinetics, muscle activity, and muscle fatigue) measures. Participants completed two unsupported, laptop-based

seated tasks, intended to represent a continuous, unsupported working condition with limited in-chair movements. We hypothesized that continuous sitting and psychosocial stress would have significant adverse effects that are reflected in both subjective and objective measures, and that there would be a correspondence between these different measures. A longer-term goal of this effort was to expand our understanding of the potential adverse consequences of continuous sitting and psychosocial stress under unsupported sitting conditions, and to provide information to develop practical interventions to reduce sitting-related discomfort and injury risks.

2 Methods

2.1 Participants

Fourteen participants (gender balanced) completed the experiment and were recruited from the local student population and community. Mean (SD) age, stature, and body mass were 24.5 (4.1) yrs, 174.5 (9.1) cm, and 72.3 (11.1) kg, respectively. After a demonstration of the experimental protocols, all participants completed an informed consent process approved by the Institutional Review Board. No participants had any self-reported musculoskeletal injuries or neurological diseases within the past year, and individuals with body mass index $>30 \text{ kg/m}^2$ were excluded due to the potential difficulties in obtaining surface EMG measures and evoking reliable muscle stimulation responses.

2.2 Experimental Design

Participants completed two tasks in separate trials that each involved continuous (40 min) use of a laptop in a seated posture without a desk (as shown in Figure 2). While some individuals may sit longer, a 40-min period was thought to represent a typical

uninterrupted period of laptop usage without a desk. Each trial involved participants sitting on a stool with no backrest. Stool height was adjusted prior to the first trial for each participant, so that both thighs were horizontal and with a 90° included knee angle. The stool height, laptop placement, and participants' postures were recorded, and the same setup was used in the second trial. To represent a continuous static sitting condition while using a laptop away from a desk (Moffet et al., 2002; Asundi et al., 2010; Gold et al., 2012), participants were instructed to complete the tasks while maintaining a relaxed kyphotic posture without large ranges of body movement or communicating with the experimenters. Participants were still allowed to alleviate any postural stresses by switching their posture between the relaxed kyphotic posture and a more upright sitting posture. Participants were also allowed to withdraw from the study at any time due to discomfort or other reason (though no participants withdrew from the study).



Figure 2. Illustration of the use of a laptop without a desk.

These tasks were done in two conditions involving different levels of psychosocial stress (low vs. higher). In the LS condition, participants engaged in relaxed internet browsing, which involved performing self-directed “surfing” (e.g., reading news, streaming video). A HS was introduced by asking participants to complete a typing task in a situation involving both additional mental workload and time pressure.

During the typing task, participants duplicated several paragraphs from a typing test program (Typing Test TQ, Giletech e.K., Germany). To induce additional mental workload, pairs of two-digit numbers were displayed between each paragraph of text, and participants were asked to mentally calculate the product or summation of these paired numbers, similar to approaches used by others (Kirschbaum et al., 1992; Hughes et al., 2007; Wang et al., 2011). To add time pressure, participants were asked to type as fast as they could, and they were offered a \$10 bonus, as an incentive, if they completed the typing task at a rate $\geq 115\%$ of their baseline speed. This baseline was measured initially during a 1-min typing task completed at an individual's normal typing speed. The order of LS and HS conditions was counterbalanced across participants, and at least four days were provided between each trial to minimize carry-over effects (Edwards et al., 1977; Jones et al., 1989).

The experimental procedures are summarized in Figure 3. The experiment started with a calibration procedure that included measures of resting and maximal voluntary muscle contractions as described earlier (Jia and Nussbaum, 2016). After a brief rest period, in which participants were allowed to move and stretch as desired, the FR angle of the trunk was measured, then SR of the lumbar extensor muscles were collected in a constrained but relaxed upright sitting position as described earlier (Jia

et al., 2015). Participants subsequently completed a 40-min continuous seated task under one of the two stress conditions.

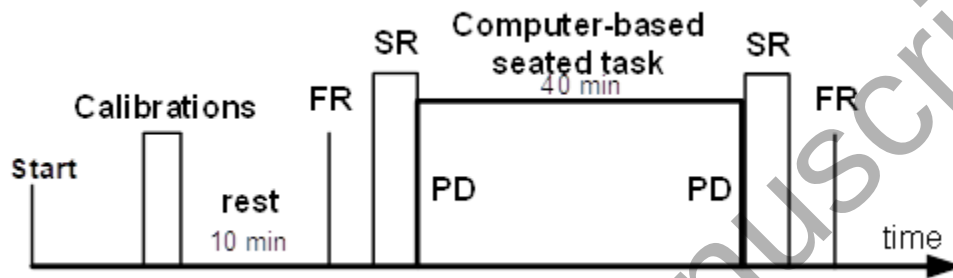


Figure 3. Illustration of experimental procedures. Calibrations: calibration trials for the EMG-based model.

At the beginning of each task, participants were also asked to rate their perceived discomfort. Upon completing each task, PD scores, SR and FR angles were collected again. Prior to the seated exposures, FR responses were collected first, since SR exposures were in a seated posture and thus could have impacted FR measures. After the seated exposures, SR responses were collected first, to minimize fatigue recovery that may have occurred if FR measures were obtained prior to these. During the 40-min tasks, EMG, trunk kinematics, and lumbosacral moments and forces were measured continuously. Data from the first minute of each 40-min task were discarded, since participants often needed this time to obtain a stable posture. Further details regarding each measure are described below.

2.3 Trunk Flexion-Relaxation Angle

Participants were asked to bend forward from an upright standing position to reach their end range-of-motion in trunk flexion (Shirado et al., 1995; Callaghan and Dunk, 2002; Solomonow et al., 2003). FR angle was then determined by the relative silence of lumbar extensor muscle activity, using procedures described earlier (Hendershot et al., 2011). For each trial, post-task FR angle was normalized to the initial value collected before the task.

2.4 Muscle Stimulation Responses

Stimulated responses of the lumbar extensor muscles were measured as an indicator of localized muscle fatigue, following the setup and protocols described in Jia et al. (2015). In brief, a dual-channel system – consisting of a stimulator (Grass S88) in series with a stimulus isolation unit (SIU5) and a constant current unit (CCU1; all from AstroMed, Inc., West Warwick, RI) – was used to electrically stimulate the lumbar extensors bilaterally at the L3 level. After a conditioning train, stimulation responses were collected (at 1000 Hz) during three, 16-second stimulation trains at 2 Hz. Post-task SR responses were normalized to initial values.

2.5 Electromyography

Surface EMG signals were collected continuously during the 40-min seated tasks. Two eight-channel, telemetered EMG systems (TeleMyo 900, Noraxon, AZ) were used to collect surface EMG bilaterally from 12 lumbar muscles, which included three lumbar extensors [EL group, ET group, and MF], and three lumbar flexors [IO, EO, and RA]. As illustrated in Figure 4, placements of disposable bipolar Ag/AgCl

electrodes (AccuSensor, Lynn Medical, MI) were derived from existing studies (McGill 1992, Mirka and Marras 1993, Cholewicki and McGill 1996, Potvin et al., 1996, Hides et al., 2008):

- EL: 3 – 4 cm lateral to the spinous process at the L3 level, parallel to the muscle fibers
- ET: 4 – 5 cm lateral to the spinous process at the T9/T10 level
- MF: 1 – 2 cm lateral to the spinous process at the L3/L4 level
- IO: 10 – 12 cm lateral to umbilicus and superior to the inguinal ligament
- EO: mid-point between 12th rib and anterior superior iliac spine
- RA: 2 – 3 cm lateral to the umbilicus



Figure 4. Front and back views showing the placement of IMUs sensors and EMG electrodes on body (labeled unilaterally)

Raw EMG signals were sampled at 1000 Hz and then amplified, band-pass filtered (20 - 400 Hz), and root-mean-square converted with a moving window of 50-millisecond duration in hardware. Root mean squared EMG signals were downsampled to 100 Hz to match the sampling rate for trunk kinematics (see below). Median power frequency was determined from the raw EMG of each muscle over 1-second windows, from the first five (i.e., 2-6 min) and the last five min. of each task; MPF was calculated using Fast Fourier Transforms as described in van Dieën et al. (1993). Mean values of EMG MPF over these 5-min samples were then obtained, and subsequently normalized to initial values.

2.6 Localized Muscle Fatigue

Localized muscle fatigue was assessed using two methods. The first method used raw EMG obtained during the trials, as described above. Mean EMG MPF obtained from the first five and the last five min of each task were calculated, and changes between these mean values were used as a fatigue index in each muscle. The second method used mean stimulation responses, with muscle fatigue indicated as a decrease in stimulation responses after vs. before the 40-min tasks (Edwards et al., 1977; Binder-Macleod and Snyder-Mackler, 1993; Johnson et al., 1995a, Jia et al., 2015).

2.7 Perceived Discomfort

Perceived discomfort, for the whole body and eight local body parts (neck, left/right shoulders, upper/lower back, buttocks and left/right thighs), was measured using a Discomfort Rating Scale (Figure 5).

Discomfort Rating Scale

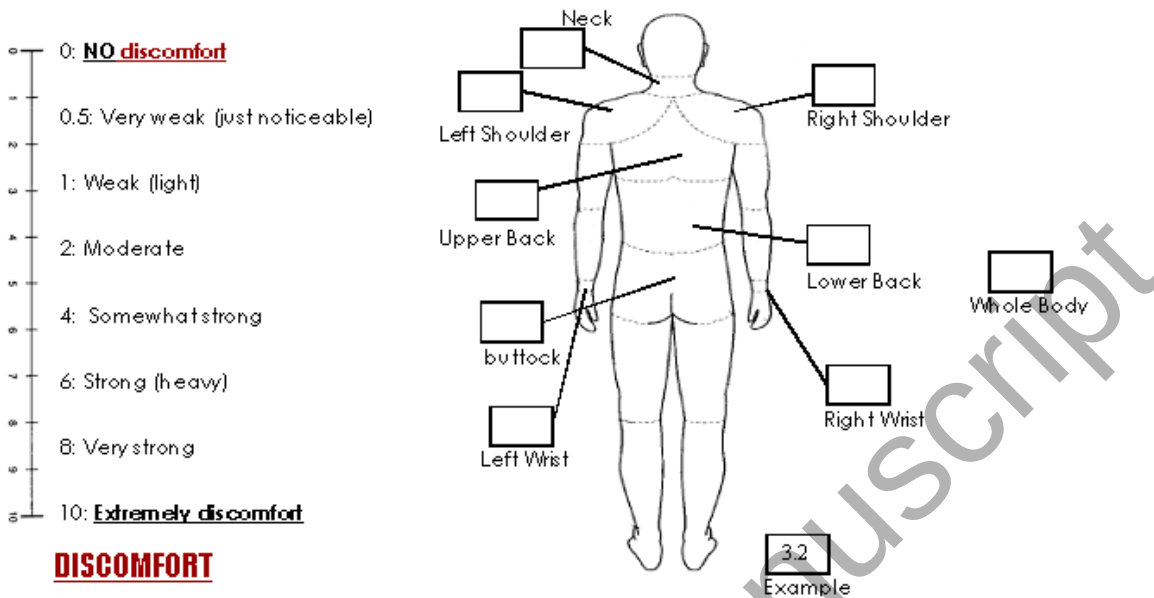


Figure 5. Methods used to obtain subjective ratings of perceived discomfort (adapted from Corlett and Bishop, 1976; Borg, 1990; and Kyung and Nussbaum, 2008)

2.8 Trunk Kinematics

Trunk angles were also collected continuously during the 40-min trials. Trunk angles were defined as the relative angles between the trunk segment and pelvis, and the latter were monitored using two inertial measurement units (IMUs: MVN, Xsens Inc., Los Angeles, CA) that were attached to the dorsal aspect of the trunk at the T10 and S1 levels (Figure 4). Both IMUs were aligned with their respective Y axes pointing superiorly, and trunk angles were calculated using Xsens software. Since participants were required to maintain a generally symmetric posture and movement in the sagittal plane was the main focus of this study, off-plane movement was assumed to be negligible. IMUs were sampled at 100 Hz and subsequently low-pass filtered at 4

Hz (4th order Butterworth; bidirectional). Mean (SD) values of trunk angles were determined from the first five and the last five min. of each task.

2.9 Lumbosacral Joint Moments and Forces

An EMG-based trunk model (Jia and Nussbaum 2016) was used to estimate lumbosacral kinetics (forces and moments). Both EMG and trunk angles were used as inputs to the model, which was calibrated to each individual. Specific model outputs were lumbosacral moments (M_{LAT} , M_{AXIAL} , and M_{FLEX}) and lumbosacral reaction forces (F_{ANT} , F_{COMP} , and F_{LAT}).

2.10 Statistical Analyses

MANOVA (using Wilks' Lambda) was used to determine the effects of continuous sitting (first vs. last five minutes), psychosocial stress, and their interaction on the dependent measures as a whole. Specific dependent measures were triaxial lumbosacral forces and moments, FR angle, trunk flexion angle, stimulation response, muscle EMG and MPF, and PD. In the event of a significant MANOVA effect, univariate ANOVAs were performed for each dependent variable where relevant. Effects were considered significant when $p < 0.05$. To assess whether there was any underlying relationship among perceived discomfort from the eight body segments, PCA was performed on the PD scores. For the same purpose, PCA analysis was also performed on EMG MPF from all measured muscles. The level of correspondence between objective measures [trunk flexion angles, compression forces, flexion/extension moments, muscle EMG MPF (principal components), and stimulation responses] and subjective measures [perceived discomfort (principal components)] was evaluated using bivariate coefficients of correlation (ρ). To assess

the consistency between the two fatigue measurement methods used, p values were also obtained between EMG median frequencies (principal components) and stimulation responses.

3 Results

MANOVA results indicated that both continuous sitting ($p<0.01$) and psychosocial stress ($p=0.04$) had significant effects on the set of dependent measures, but there was not a significant interaction effect ($p=0.90$). Results of univariate ANOVAs (Table 1) indicated that continuous sitting had significant main effects on a number of dependent measures, which included trunk flexion angles, stimulation responses, MPF from most muscles, and perceived discomfort in all body regions evaluated. In contrast, psychosocial stress had relatively few significant effects.

Table 1. Summary of ANOVA results for the main effects of continuous sitting and psychosocial stress. Significant effects are highlighted in bold text.

		Continuous Sitting		Psychosocial Stress	
		$F_{(1,1057)}$	p	$F_{(1,1057)}$	p
Lumbosacral Moments and Forces	F_{ANT}	0.09	0.76	1.63	0.21
	F_{COMP}	0.87	0.36	6.89	<0.01
	F_{LAT}	20.09	<0.01	14.41	<0.01
	M_{LAT}	0.67	0.42	8.53	0.01
	M_{AXIAL}	0.28	0.60	0.74	0.40
	M_{FLEX}	0.12	0.74	1.78	0.19
Trunk Flexion Angle		8.32	<0.01	1.09	0.30
FR Angle		0.49	0.49	2.83	0.10
SR		4.01	0.05	0.11	0.74
EMG Median Frequency	LMF	2.16	0.15	0.07	0.79
	LEL	7.65	<0.01	0.15	0.71
	LET	5.36	0.02	1.02	0.32
	LEO	5.46	0.03	0.11	0.75
	LIO	6.51	0.02	0.25	0.62
	LRA	2.71	0.12	2.59	0.12
	RMF	10.58	<0.01	0.17	0.68
	REL	6.25	0.02	3.11	0.09
	RET	4.34	0.04	3.02	0.09
	REO	0.03	0.86	0.13	0.73
	RIO	6.38	0.02	0.94	0.34
	RRA	5.10	0.03	1.23	0.74
Perceived Discomfort	Neck	35.31	<0.01	1.01	0.32
	L Shoulder	22.59	<0.01	0.01	0.94
	R Shoulder	32.45	<0.01	0.09	0.77
	Upper Back	106.53	<0.01	1.94	0.17
	Lower Back	39.64	<0.01	0.03	0.86
	Buttock	27.86	<0.01	0.32	0.57
	L Wrist	5.68	0.02	1.71	0.20
	R Wrist	11.85	<0.01	0.48	0.49
Whole Body		104.07	<0.01	1.02	0.32

FR, SR, and EMG MPF results are summarized in Figure 6. Although not significant, FR angle generally increased at the end of each task with large between-subject variation. Signs of muscle fatigue were found from both fatigue measurement methods. Continuous sitting resulted in a significant decrease in SR and significant declines in MPF from all muscles except the LMF, LRA, and REO. No main or

interaction effects of psychosocial stress were found for stimulation response, EMG MPF, or FR angle. Though not significant, both fatigue measurements indicated more fatigue in the HS condition.

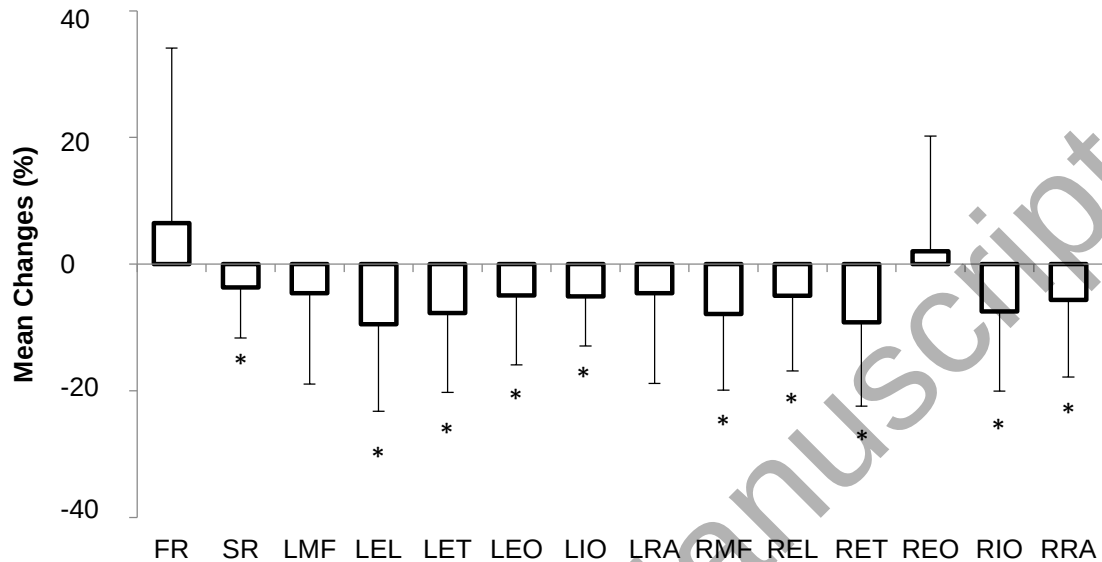


Figure 6. Illustration of mean changes (after vs. before task) in normalized flexion-relaxation angles (FR), stimulation responses (SR), and EMG median frequencies. All measures were normalized to initial values (i.e., 0% = no change over time). The * symbol indicates a significant effect of continuous sitting, and error bars indicate standard deviations.

PD results are summarized in Figure 7. Discomfort for the whole body and for each of the eight local body parts increased significantly by the end of the 40-min trials. The highest level of perceived discomfort was found in the upper back, followed by the lower back and whole body. No main or interaction effects of psychosocial stress were found on PD measures.

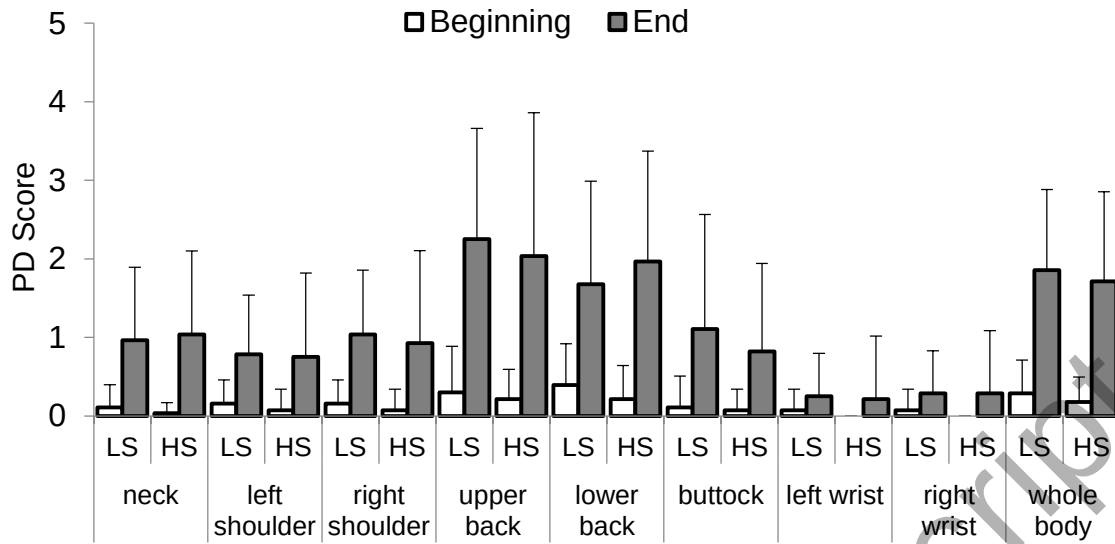


Figure 7. Mean (SD) values of perceived discomfort for each body part. All PD scores significantly ($p < 0.01$) increased at the end of the task. Error bars indicate standard deviations.

Across all experimental trials, mean levels of muscle activity were typically lower than 10% of MVC) and the highest levels of muscle activity were observed for the bilateral ETs. No significant ($p > 0.10$) differences in mean muscle activity levels were found between the two psychosocial stress conditions. However, activation levels of the trunk extensors were typically higher in the HS condition (Figure 8). Activation levels of the trunk flexors, in contrast, were generally lower in the HS condition excepting the left IO.

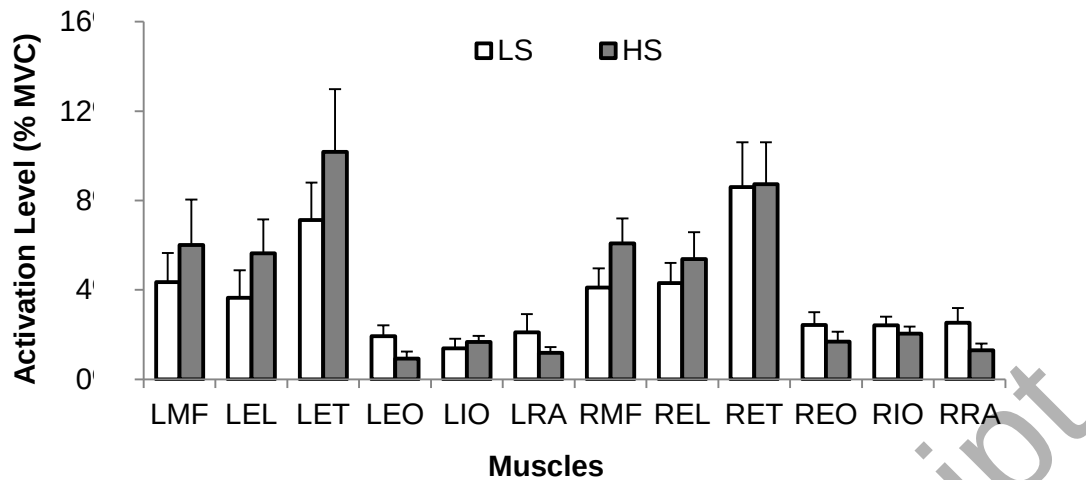


Figure 8. Summary of muscle activation levels between the two stress conditions.

Error bars indicate standard deviations.

Across all trials, mean (SD) lumbosacral moments in lateral bending, axial rotation, and flexion were 1.7 (1.4), 2.4 (2.5), and 19.9 (8.2) Nm, respectively. Respective mean (SD) levels of lumbosacral forces were 68.1 (43.5), 771.5 (205.6), and 24.0 (22.1) N for anterior shear, compression, and lateral shear. Continuous sitting significantly increased lateral shear forces, from 18.0 (18.0) N to 29.9 (24.4) N. The higher psychosocial stress condition increased compression forces [from 723.6 (164.7) N in LS vs. 819.3 (232.7) N in HS], and lateral bending moments [from 1.6 (1.1) Nm in LS vs. 1.8 (1.6) Nm in HS], but decreased lateral shear forces [from 25.83 (26.3) N in LS vs. 22.1 (17.1) N in HS]. Continuous sitting also resulted in higher trunk flexion angles, from 12.5 (11.0)° initially to 19.2 (6.8)° at the end of the tasks.

For both perceived discomfort scores and EMG MPF, principal component analyses revealed one dominant PC for each, and which accounted for > 38% of the variance in EMG MPF and > 67% of the variance in perceived discomfort scores (Table 2).

The dominant component, EPC1, varies together with all the EMG MPF, which can be viewed as a measure of overall level of fatigue. Likewise, the dominant component PPC1 of perceived discomfort scores can be viewed as representing the overall level of perceived discomfort.

Table 2. Principal component coefficients for EMG MPF and PD scores.

		Left		Right		Upper	Lower			Left	Right	Whole
PD	Neck	Shoulder		Shoulder		Back	Back	Buttock		Wrist	Wrist	Body
PPC1	0.3	0.36		0.35		0.32	0.32	0.33		0.34	0.35	0.33
EMG	LMF	LEL	LET	LEO	LIO	LRA	RMF	REL	RET	REO	RIO	RRA
EPC1	0.15	0.37	0.12	0.37	0.19	0.40	0.27	0.35	0.17	0.20	0.31	0.38

As shown in Figure 9, no strong or significant relationships were found between objective and subjective measures. Although not significant, PPC1 was positively associated with trunk flexion angle, compression forces, and flexion moments, and was negatively associated with stimulation responses and EPC1. A positive, though not significant, relationship was found between EPC1 and stimulation responses.

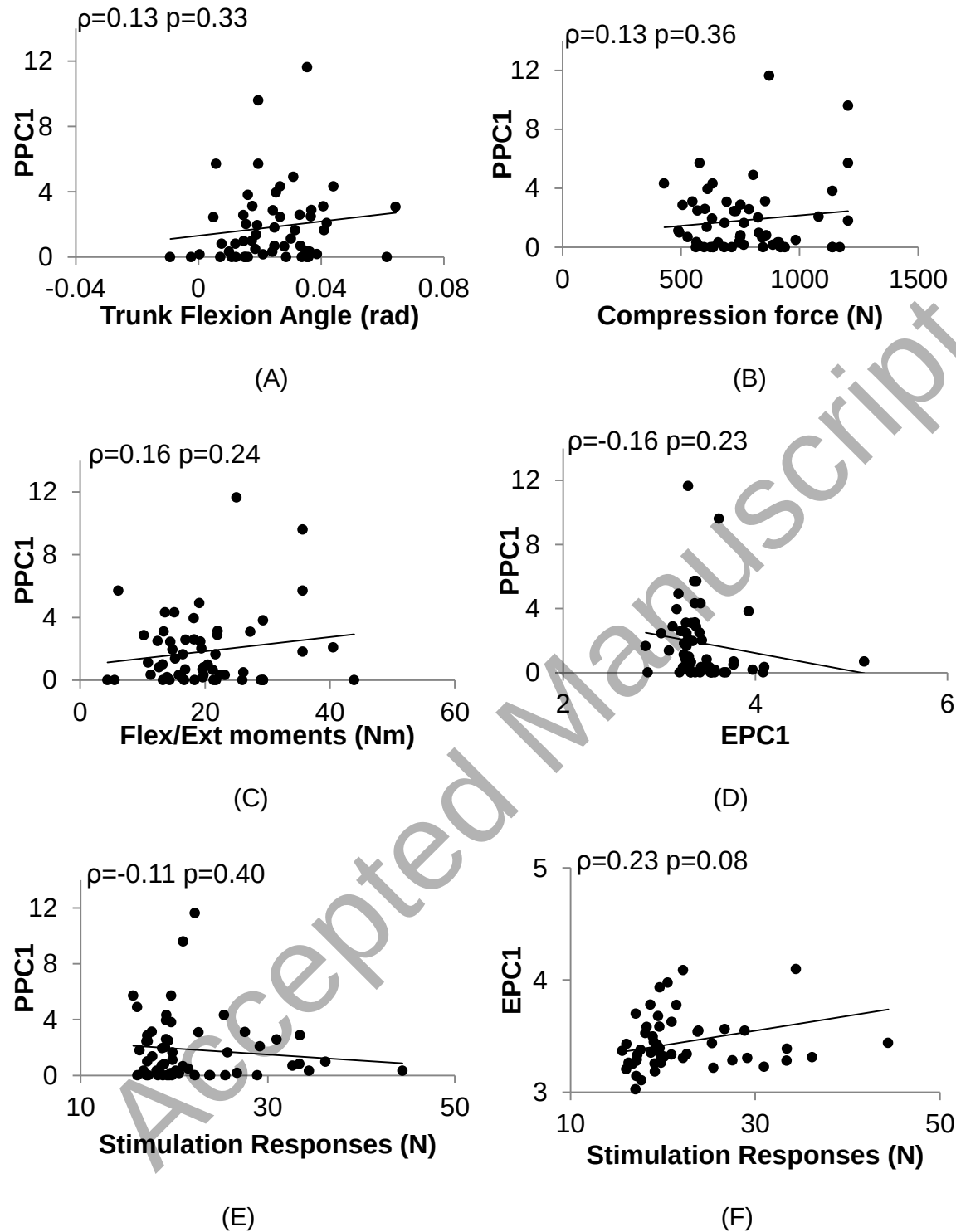


Figure 9. Relationships between subjective (PPC1) and objective measures [lumbar angle (A), compression force (B), flexion moment (C), EPC1 (D) and stimulation responses (E)], as well as between stimulation responses and EPC1 (F).

4 Discussion

The objective of this study was to determine the effects of continuous sitting and psychosocial stress, in the context of unsupported sitting, on lumbosacral kinematics and kinetics, perceived discomfort, and localized muscle fatigue. Only the main effects of continuous sitting and psychosocial stress on selected dependent measures were significant, which suggests that the effects of different level of stress were consistent across the 40-min sitting periods examined. It is unclear, though, if this lack of an interaction effect will generalize to a larger range of exposure conditions or longer sitting durations.

Continuous sitting significantly increased the magnitude of lateral shear forces ($p < 0.01$). As noted earlier, absolute values were analyzed, though these forces varied over a wide range (from - 55.17 N to 111.01 N). The tasks here were completed in largely symmetrical postures, for which the major trunk rotations and major lumbosacral kinetics were in the sagittal plane. Previous evidence indicates that participants have less effective motor control, especially when muscles are fatigued, in the coronal and transverse planes when performing a task mainly in the sagittal plane, and which results in substantially different muscle recruitment patterns for identical tasks (Parnianpour et al., 1988). As such, the observed changes in predicted lateral shearing forces may have been caused by underlying changes in muscle recruitment during continuous sitting.

While the impacts of psychosocial stress were significant for only a limited number of the current dependent measures, the higher psychosocial stress condition (HS) did increase compression forces by 12%. Previous work has indicated that increased

psychosocial stress could contribute to increased muscle activation, especially during conditions involving relatively low physical demands (Marras et al., 2000; Hughes, 2004; Hughes et al., 2007); such increased muscle activity was found here (Figure 8), though the differences were not significant. These increased muscle activations, however, likely contributed to the increased predicted compression forces (Cholewicki et al., 1995; Granata and Marras, 2000).

There were significant increases in trunk flexion angles at the end of both tasks. Participants were asked to maintain a static, kyphosed sitting posture, with minimal posture adjustments or in-chair movement. Both creep and fatigue in the low back can develop with static trunk flexion (Kazarian, 1975; Keller et al., 1987; Adams and Dolan, 1996; Kanlayanaphotporn et al., 2002), and creep may occur after only five minutes of static flexion (Adams and Dolan, 1996; Kanlayanaphotporn et al., 2002). FR angle is often used as an indicator of creep deformation in the spinal structure under flexion exposures, with results from such studies indicating that continuous trunk flexion increases FR angle (Olson et al., 2004; Olson et al., 2006; Shin et al., 2009). Here, the increase in trunk flexion angles indicated a potential trunk deformation over the 40 min of continuous sitting, which was further supported by the increase (albeit not statistically significant) in FR angles.

Localized fatigue was measured using both muscle stimulation and EMG MPF. Continuous sitting resulted in significant decreases in stimulated responses and declines in EMG MPF for several flexor and extensor muscles at the end of the 40-min periods in both psychosocial stress conditions. These decreases suggest that localized muscle fatigue did occur as a result of completing the continuous,

unsupported seated task in a kyphosed sitting posture. Some of the inconsistencies between muscles may be attributed to functional differences between the trunk flexors and extensors. During continuous sitting, trunk extensors are usually considered as postural muscles, and which are required to be continuously activated to stabilize the sitting posture (van Dieën et al., 2001; Mork and Westgaard, 2005; Mork and Westgaard, 2009). Consistent with this functional muscle role, we observed declines in EMG MPF among three trunk extensors (EL, ET, and MF). The observed decrease in EMG MPF in the bilateral IO also supports the role of the IO muscle in stabilizing the trunk during sitting, consistent with some existing evidence (Snijders et al., 1995).

The correspondence between two distinct fatigue measurement methods was determined. Only a weak positive correlation was evident between EMG MPF and stimulation responses (Figure 9E), though with indications in both responses of localized fatigue developing by the end of the continuous seated tasks. Consistency between these different measures supports that fatigue indeed developed during these tasks, though the lack of a strong correspondence between them suggests that they may be reflecting different underlying phenomena and that both approaches have potential value in assessing muscle fatigue during continuous sitting.

Perceived discomfort also significantly increased at the end of 40 min of continuous sitting. A continuous static sitting posture, such as that imposed by the non-desk setup in this study, has been shown to lead to discomfort in the neck, shoulders, and back (Grandjean et al., 1983; Stokes et al., 1987; Gay et al., 2006), and which increases over time during uninterrupted tasks at work (Marras et al., 1995; Ostensvik

et al., 2009). To minimize potential confounding effects of in-chair movement, participants here maintained a kyphosed sitting posture, as one example of a sitting posture adopted in non-desk conditions, and which can increase intradiscal pressures (Wilcox and Keselman, 2003), stretch the posterior spine ligaments (Owen et al., 2010), and restrict nutrient supplies to nerves (Hallal et al., 2012). These factors may have contributed to the observed increases in discomfort.

While subjective measures (e.g., perceived discomfort) provided direct assessment of sitting discomfort, objective measures can potentially provide unbiased estimations that are related to sitting discomfort (Lee et al., 1993; Kyung et al., 2008). Support for objective measures as useful additions to subjective measurements of sitting discomfort would be provided if a good correlation existed between these two types of measures. Several studies have been completed to identify the relationship between subjective measures (e.g., perceived discomfort) and a range of objective measures including trunk posture, EMG, and interface pressure (Eklund and Corlett, 1987; Lee et al., 1993; Salewytch and Callaghan, 1999; Vergara and Page, 2000). Here, the first principal component of PD scores was used as a representation of overall perceived discomfort across all measured body parts and the whole body, and correlations were obtained between PPC1 and several objective measures (trunk flexion angle, muscle EMG MPF and stimulation responses, Figure 9). Although no statistically significant ($p > 0.23$) correlations were found between subjective and objective measures, trunk flexion angle was positively associated with perceived discomfort, specifically with higher trunk flexion angles corresponding with higher perceived discomfort, a relationship consistent with previous evidence (Eklund and Corlett, 1987). Declines in both fatigue measures (EPC1 and stimulation response)

were also found to be associated with an increase in perceived discomfort. This association suggests that participants may have perceived higher discomfort, at least in part, as a result of localized muscle fatigue, as has been described earlier (Rohmert and Luczak, 1978; Zhang et al., 1996).

Four potential limitations of this work need to be acknowledged. First, the effects of continuous sitting and psychosocial stress were evaluated under a constrained sitting condition. Each experimental task was completed with participants sitting in a single, fixed posture, and with minimal in-chair moments. In realistic sitting situations, participants are free to make postural adjustments or take microbreaks to reduce body discomfort (Salewytch and Callaghan, 1999; McLean et al., 2001). Therefore, the results of the current study may have limited ability to generalize to the broader use of a laptop or other mobile device in various unsupported sitting conditions. Future work is needed to test the effects of continuous sitting and psychosocial stress under more complex task conditions that include a wider range of trunk movements and sitting postures. Second, the duration of the continuous task was determined based on the results from our pilot studies, where muscle fatigue was observed after 30 min of quiet sitting. There is a lack of evidence, to our knowledge, regarding typical exposure durations for laptop use without a desk, so the applicability of the current work, in terms of the duration of actual use, is unknown. In addition, 40 min of continuous sitting may not be of sufficient duration to detect potential time-dependent changes in passive trunk components (via viscoelastic changes). Additional study is needed to identify the common exposure durations of sitting under different conditions and a longer duration could be also expected to detect the time-dependent changes associated with passive and active trunk components. Third, only two

psychosocial stress levels were evaluated in this study, which may not reflect the spectrum of psychosocial stress situations. Including more stress conditions may help to better identify the effects of psychosocial stress over time. Fourth, while significant increases in whole-body and localized discomfort were found as a result of the 40-min. seated exposures, the final values were relatively low overall (Figure 7) and it is unclear if these values represent practically-meaningful level, and the lack or more substantial discomfort may have precluded identifying stronger correlations between discomfort and objective measures (Figure 9).

In summary, unsupported continuous sitting was found to have important impacts on lumbosacral kinematics, and to lead to the development of muscle fatigue and perceived discomfort. The current findings suggest that sitting for moderate continuous periods (i.e., 40 min.) can lead to higher body discomfort, internal spinal loads, and localized fatigue, which could further increase risks of developing LBP. Though more limited effects of psychosocial stress were found, the higher-level stress imposed here did lead to an increase in lumbosacral compression. These outcomes were evident after 40 min of sitting under experimental conditions involving laptop use without a desk, which could be of particular concern for some conditions of mobile device use (e.g., in an airport). This study highlighted the potential influence of sitting-related moderating factors on individual behaviors and provided evidence to facilitate a better understanding of sitting-related risks. Further research is still needed, however, to assess the effects of continuous sitting and psychosocial stress in more complex occupational and non-occupational sitting conditions through more detailed analysis, such as estimation of time-sensitive lumbosacral loads using advanced spine model.

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